

Education

Bachelor of Science in Applied Mathematics & Computer Science

University of Delaware, Newark, DE

Honors College | Expected May 2027

Work Experience

- **Data Science Intern — Victory Capital Management (Python, SQL, K-Means Clustering)**

- Pulled user-level data from Amazon Redshift with SQL to build datasets for K-Means clustering, enabling actionable insights into customer behavior
- Diagnosed and resolved data and modeling issues in a production K-Means pipeline, including low-signal features, misinterpreted clusters, and inconsistent behavioral definitions
- Redesigned feature engineering pipeline, increasing silhouette scores and producing more interpretable clusters, helping the marketing team target user segments more effectively
- Streamlined cluster usability for downstream analytics, improving decision-making for marketing campaigns

Selected Machine Learning Projects

- **Sports Analytic Predictions (Python, XGBoost):**

- Built end-to-end XGBoost pipeline on real NFL rush yard game data, achieving actionable predictive insights for rushing yards and over/under outcomes
- Engineered rolling workload, efficiency, and matchup features from historical game data
- Achieved RMSE of 27.8 yards and R^2 of 0.46 on one season of out-of-sample data; classification model achieved ROC-AUC of 0.626 over a two-month evaluation window

- **Trading Bots (Python, AWS, XGBoost):**

- Designed and deployed Python-based algorithmic trading bots on AWS using live WebSocket market data for real-time SPY options execution
- Integrated XGBoost predictive model to drive live trading decisions, producing +4% expected value in backtested real-market scenarios
- Implemented automated execution logic and risk controls, including trailing stops and position sizing

Technical Skills

- **Languages:** Python, SQL, C++, C, JavaScript, TypeScript
- **Machine Learning / Data Science:** scikit-learn (K-Means, model evaluation), XGBoost, feature engineering, statistical modeling, exploratory data analysis, hypothesis testing
- **Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly
- **Cloud & Infrastructure:** AWS (S3, Redshift, ECS, ECR), Docker, Firebase
- **Developer Tools & Systems:** Git, Bash, Jupyter Notebook, Visual Studio Code, WebSocket-based streaming

University Activities

- **Paid Instructional Assistant – General Computer Science for Engineers**

- Lead lectures, assisting engineering students new to computer science in understanding core programming concepts
- Provide one-on-one support to students in Python programming, troubleshooting code, and reinforcing best practices